NEWTON SCHOOL

PRE-ALGEBRA | ANSWER KEY

Homework Answer Key: Parallel Lines, Transversals & Polygon Angles

Student: Natalie Yoo **Date:** 2026-05-14

Standard: *CCSS.MATH.CONTENT.8.G.A.5 — Angle relationships with parallel lines and transversals; interior angle sum of polygons.*

Part A — Name the Angle Pair (Q1–5)

Problem 1

$\angle 3$ and $\angle 7$ — Angle pair name: ___________

Alternate Interior Angles | Parallel Lines

$\angle 3$ is below line ℓ (inside the parallel lines), on the right side of the transversal.

$\angle 7$ is above line m (inside the parallel lines), on the right side of the transversal.

They are on *opposite* sides of the transversal, both *between* the two parallel lines → **Alternate Interior**.

Alternate Interior Angles

Problem 2

$\angle 1$ and $\angle 5$ — Angle pair name: ___________

Corresponding Angles | Parallel Lines

$\angle 1$ is in the upper-left position at the upper intersection.

$\angle 5$ is in the upper-left position at the lower intersection.

Same position at each intersection → **Corresponding**. (They are equal when lines are parallel.)

Corresponding Angles

Problem 3

$\angle 4$ and $\angle 5$ — Angle pair name: ___________

Co-Interior (Consecutive Interior) Angles | Supplementary

$\angle 4$ is on the left of the transversal, below line ℓ.

$\angle 5$ is on the left of the transversal, above line m.

Both are between the parallel lines on the *same* side of the transversal → **Co-Interior (Consecutive Interior)**. They add up to 180°.

Co-Interior (Consecutive Interior) Angles

Problem 4

$\angle 2$ and $\angle 6$ — Angle pair name: ___________

Corresponding Angles

$\angle 2$ is upper-right at the top intersection; $\angle 6$ is upper-right at the bottom intersection.

Same corner, different intersection → **Corresponding**.

Corresponding Angles

Problem 5

$\angle 1$ and $\angle 7$ — Angle pair name: ___________

Alternate Exterior Angles

$\angle 1$ is above line ℓ (outside the parallel lines), upper-left.

$\angle 7$ is below line m (outside the parallel lines), lower-right.

Both are *outside* the parallel lines on *opposite* sides of the transversal → **Alternate Exterior**.

Alternate Exterior Angles

Part B — Solve for x (Q6–10)

Problem 6

Corresponding angles: 75° and $(11x - 2)^\circ$. Solve for x.

Corresponding Angles | Equal

Corresponding angles are equal when lines are parallel:

$11x - 2 = 75$

$11x = 75 + 2 = 77$

$x = 77 \div 11 = 7$

$\boxed{x = 7}$

Problem 7

Alternate interior angles: $(3x + 15)^\circ$ and 51°. Solve for x.

Alternate Interior Angles | Equal

Alternate interior angles are equal:

$3x + 15 = 51$

$3x = 51 - 15 = 36$

$x = 36 \div 3 = 12$

$\boxed{x = 12}$

Problem 8

> Co-interior angles sum to 180°: 110° and $(7x+5)^\circ$. Solve for x.

Co-Interior Angles | Supplementary

Co-interior (consecutive interior) angles add to 180°:

$110 + (7x + 5) = 180$

$115 + 7x = 180$

$7x = 180 - 115 = 65$

$x = 65 \div 7$ — Wait, let's check: $7 \times 9 = 63$, $7 \times 10 = 70$. Hmm, let's re-verify: $7x = 65$. Actually $x = 65/7$. Recheck setup: $110 + 7x + 5 = 180 \to 7x = 65 \to x = \frac{65}{7} \approx 9.3$. Clean integer check: this is correct as given.

Re-approach for clean answer: $7x = 180 - 115 = 65 \to x = \frac{65}{7}$. Note: the problem yields $x = \frac{65}{7}$. Verify: $7(\frac{65}{7}) + 5 = 65 + 5 = 70$; $110 + 70 = 180$. ✓

$$\boxed{x = \frac{65}{7} \approx 9.3}$$

Problem 9

> Alternate exterior angles: $(5x-10)^\circ$ and 60°. Solve for x.

Alternate Exterior Angles | Equal

Alternate exterior angles are equal:

$5x - 10 = 60$

$5x = 60 + 10 = 70$

$x = 70 \div 5 = 14$

$$\boxed{x = 14}$$

Problem 10

> Co-interior angles: $(4x+12)^\circ$ and $(2x+24)^\circ$ sum to 180°. Solve for x.

Co-Interior Angles | Two-Step

$(4x+12)+(2x+24)=180$

$6x+36=180$

$6x=180-36=144$

$x=144\div 6=24$

Check: $(4(24)+12)+(2(24)+24)=108+72=180\checkmark$

$\boxed{x=24}$

Part C — Quadrilaterals (Q11–13)

Problem 11

> A quadrilateral has all four sides equal and all four angles equal to 90°. Most specific name?

Quadrilateral Classification | Square

All 4 sides equal → could be rhombus or square.

All 4 angles = 90° → rectangle or square.

Both conditions together → **Square** (most specific name).

Square

Problem 12

> A quadrilateral has angles 80°, 100°, 80°. Find the missing fourth angle.

Angle Sum = 360° | Missing Angle

Sum of all four angles in any quadrilateral = 360°.

$80 + 100 + 80 + x = 360$

$260 + x = 360$

$x = 360 - 260 = 100$

Missing angle = 100°

Problem 13

> Exactly one pair of parallel sides; angles 65°, 115°, 65°, 115°. Type?

Trapezoid | Quadrilateral

Exactly one pair of parallel sides → **Trapezoid**.

Co-interior angles on the same leg add to 180°: $65^\circ + 115^\circ = 180^\circ$ ✓ (confirms the parallel sides).

Trapezoid

Part D — Polygon Basics (Q14–17)

Problem 14

> Stop sign: 8 equal sides, 8 equal angles. Name? Regular or Irregular?

Octagon Regular

8 sides → **Octagon**.

All sides equal AND all angles equal → **Regular**.

Regular Octagon

Problem 15

> Is a circle a polygon? Yes or No?

Polygon Definition

A polygon must be a closed figure made of *straight* line segments.

A circle has no straight sides — it is a curved shape.

No — a circle is not a polygon because it has no straight sides.

Problem 16

> 10 sides, one interior angle > 180°. Name and convex/concave?

Decagon Concave

10 sides → **Decagon**.

An interior angle > 180° means the shape "caves in" — it is **Concave**.

Concave Decagon

Problem 17

> 12 sides, sides not all the same length. Name and regular/irregular?

Dodecagon Irregular

12 sides → **Dodecagon**.

Sides not all equal → **Irregular**.

Irregular Dodecagon

Part E — Interior Angle Sum (Q18–20)

Problem 18

Find the sum of interior angles of an 11-gon (hendecagon).

Interior Angle Sum Formula | $(n-2) \times 180^\circ$

Formula: $(n-2) \times 180^\circ$

$n = 11$: $(11-2) \times 180 = 9 \times 180$

$9 \times 180 = 1{,}620$

Sum = $1{,}620^\circ$

Problem 19

Find the sum of interior angles of a 25-gon.

Interior Angle Sum Formula

$n = 25$: $(25-2) \times 180 = 23 \times 180$

$23 \times 180 = 23 \times 100 + 23 \times 80 = 2{,}300 + 1{,}840 = 4{,}140$

Sum = $4{,}140^\circ$

Problem 20

A regular octagon has 8 equal angles. What is the measure of each interior angle?

Regular Polygon | Each Interior Angle

Step 1 — Find the total sum: $(8-2) \times 180 = 6 \times 180 = 1{,}080^\circ$

Step 2 — Divide equally: $1{,}080 \div 8 = 135^\circ$

Each interior angle = 135°

Quick Reference — Angle Pair Summary

Corresponding: same position, different intersection → *equal*

Alternate Interior: between lines, opposite sides → *equal*

Alternate Exterior: outside lines, opposite sides → *equal*

Co-Interior (Consecutive Interior): between lines, same side → *add to 180°*

Vertical: across the intersection point → *equal*

Linear Pair: adjacent, form a straight line → *add to 180°*